CHE 205 Final Exam May 8th, 2020

NAME: ________________________

- Exam is open-book and open-note. You are allowed to use internet resources, however, you are NOT allowed to talk to, text or email anybody during the exam.
- You need to submit your excel spreadsheet on BB before 1pm. Make sure you save your file frequently and save as .xlsm to be able to save your VBA codes. If you do not save as .xlsm, I will not be able to see your VBA code and you will lose points on those problems. Please write step-by-step explanations on a paper and submit the copy of that on BB with your excel sheet.
- Make sure you keep a copy of your work on your computer.
- Highlight your answers and make sure to label constants, initial values and equations. I need to follow your logic.
- Save each problem as different sheet in the same excel file. Screenshot your VBA code and paste it into excel.
- Exam is 125 points in total. Good luck!

Question 1 (25 points):
Consider the following equation:

$$\sqrt{\frac{1}{x}} = a * \ln\left(b + \frac{c}{x}\right)$$

where $a = 10,\ b = 8, and\ c = 0.01\ and \ln is\ natural\ log. 0.0001 < x < 0.005$
(a) To find the root of above equation, you need to convert that to a function. Plot the function whose root you are trying to find.
(b) Based on your results in part a, make two good initial guesses, x_1 and x_2. Using these initial guesses and Secant method, solve for x.
(c) Use VBA code from practice final exam. Input your initial guesses from part b and the tolerance, and output your solution for x, total number of iterations, and the error at each iteration.

Question 2 (25 points):
Reaction kinetics for a bioreactor is given by:

$$\frac{ds}{dt} = r_1 s - \frac{r_1}{r_2} s^2$$

$$\frac{dx}{dt} = r_3 s$$

$$\frac{dy}{dt} = r_4 x$$

s = substrate concentration (M/L^3)
x = biomass concentration (M/L^3)

y = byproduct concentration ((M/L^3)

Rate constants are determined by experiments and found to be $r_1 = 12.9, r_2 = 0.89, r_3 = 1.67, and\ r_4 = 0.75$. Initial conditions are s(0) = 0.05, x(0) = 0, and y(0) = 0.

(a) List the methods that you can use to solve this problem.

(b) Solve this problem with forward-Euler method in spreadsheet between $0 \le t \le 2$ with $\Delta t = 0.01$. Plot s, x, and y versus time. (Not VBA programming)

(c) Using VBA program from practice final exam, solve this problem using 4th-order RK method. Plot s, x, and y vs time.

Question 3 (25 points):

Consider following heat equation with a heat sink (the second term on the right-hand side).

$$\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2} - 2T$$

This equation is subject to the following initial and boundary conditions

$$T(x, t = 0) = 0, \qquad \frac{\partial T(x = 0, t)}{\partial x} = -2, \qquad T(x = 1, t) = 0,$$

where T is temperature, $0 \le t \le 0.1$ is the time variable and x is the spatial coordinate. Use $\Delta x = 0.05, and\ \Delta t = 0.005$.

(a) List the methods that you can use to solve this equation.

(b) Using the Crank-Nicholson method and VBA program from the practice final exam, solve this equation and plot T (x, t=0.1) versus $0 \le x \le 1$.

Question 4 (25 points):

Heat transfer in a fin with variable thermal conductivity follows the equation below.

$$(1 + x)\frac{d^2T}{dx^2} + L\frac{dT}{dx} = \alpha(T - T_a),$$

where $0 \le x \le 1,\ T = T(x)$ is the temperature at point x, L = 0.25 ft is the length of the fin, α = 6.5 and T_a = 0.19. Temperature is fixed on the left, T(0) = 1, and is following boundary condition on the right:

$$\frac{dT}{dx} = -0.05(T - T_a), at\ x = 1,$$

(a) List methods to solve this problem.

(b) Solve this problem with forward Euler together with "shooting method" in a spreadsheet. For discretization, use $\Delta x = 0.05$. Plot T(x) against x. (This is not a VBA programming)

Question 5 (25 points):

Consider the data shown below.

v	y
0	1.766
0.25	2.478
0.5	3.690
0.75	6.397
1	6.649

(a)Plot data in Excel and add a quadratic polynomial trendline.

(b)Calculate c_1, c_2, and c_3 by hand in Excel spreadsheet using regression analysis.

(c)Calculate R^2 value by hand in Excel spreadsheet and compare with trendline values.